	Faculty Department: CSE **Academic Year:2025-2026**	**Submission Date:** **09/02/2026** **Max. Marks: 40**
	B.E & B.Tech. **Specialization: CS and AD**	
	Student Department: CSE **Semester: IV** **Assignment No.: 1.2**	
	Sub. Code & Name: CS22401-Operating Systems	

SET 1

Q. No	Question / Task / Activity	CO*	RBT Level	Marks
1	The table given below lists the processes, corresponding Burst time, Arrival time and Priority. Perform CPU Scheduling for various scheduling algorithms, draw the process Gantt chart and calculate the following for each category. (Note: all time units given in Nano seconds) a. Finishing time b. Waiting time c. Turnaround time d. Average waiting time e. Average Turnaround time Category of scheduling algorithms: **FCFS** a. Without Arrival time b. With Arrival time **SJF** c. Without Arrival time d. Preemptive (With Arrival time) e. Non-Preemptive (With Arrival time) **PRIORITY** f. Without Arrival time g. Preemptive (With Arrival time) h. Non-Preemptive (With Arrival time) **ROUND ROBIN (Time Slice – 2ns)**	2	AN	10
2	In a supermarket customers are waiting for billing their items they bought. Time taken for each item billing is 1 unit. Customer John has 12 items and Alice has 3 items for billing and they arrived at t=0, whereas Boby arrived at t=1 and she has 7 items for billing. Shillon arrived at t=2 who has 2 items for billing. The billing system has the capability of halting the process of a current customer and can service the new customer. Do a comparative analysis of FCFS, SJF (Non-Pre-emptive, Pre-emptive), Priority (alphabetical ordering of names), round robin (time slice=2 units) and provide a suitable advice to the owner of the shop which method can be adopted for increasing the customer base.			10

Process	Arrival Time	Burst Time	Priority
P1	0	2	2
P2	1	5	6
P3	7	10	1
P4	5	8	4
P5	6	3	3
P6	8	9	5

3	Consider a case study of your own and apply SJF and priority scheduling and calculate the average TAT, FT and WT. (Apply your own knowledge on CPU scheduling). Burst time will be the last three digits of your registration number thereby increase by five for the four processes. Note: If your registration number is 2127240501001 then the burst time for Process 1 is 001 and thereby increase by 5 for the Process 2, Process 3 and Process 4.			**20**
			Total Marks	**40**

SET 2

<table>
<tr><th>Q. No</th><th>Question / Task / Activity</th><th>CO*</th><th>RBT Level</th><th>Marks</th></tr>
<tr><td>1</td><td>I) The table given below lists the processes, corresponding Burst time, Arrival time and Priority. Perform CPU Scheduling for various scheduling algorithms, draw the process Gantt chart and calculate the following for each category. (Note: all time units given in Nano seconds)

a. Finishing time
b. Waiting time
c. Turnaround time
d. Average waiting time
e. Average Turnaround time.
Category of scheduling algorithms:
FCFS
a. Without Arrival time
b. With Arrival time
2. SJF
a. Without Arrival time
b. Preemptive (With Arrival time)
c. Non-Preemptive (With Arrival time)
PRIORITY
d. Without Arrival time
e. Preemptive (With Arrival time)
f. Non-Preemptive (With Arrival time)
ROUND ROBIN (Time Slice – 3ns)
<table>
<tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr>
<tr><td>P1</td><td>0</td><td>3</td><td>4</td></tr>
<tr><td>P2</td><td>1</td><td>6</td><td>2</td></tr>
<tr><td>P3</td><td>4</td><td>4</td><td>5</td></tr>
<tr><td>P4</td><td>5</td><td>5</td><td>1</td></tr>
<tr><td>P5</td><td>6</td><td>2</td><td>3</td></tr>
<tr><td>P6</td><td>8</td><td>4</td><td>6</td></tr>
</table></td><td rowspan="2">2</td><td rowspan="2">AN</td><td>10</td></tr>
<tr><td>2</td><td>A printing machine is common for 5 employees in a firm. Worker1 has 10 sent pages for printing at t=10.00 Worker2 and Manager sent 5pages ,4 pages respectively at t=10.01 for printing. Worker3 has sent 100 pages at t=10.02 and Supervisor has sent 12 pages at 10.05 for printing. The jobs are placed in the printer buffer and the scheduler picks up the job and sends for printing. Implement the following scheduling for printing the jobs. Time to print a page is 1sec.
1.FCFS
2.SJF
3.SRTF
4.Round Robin (t= 2 secs)
5.Non-pre-emptive priority (order of priority Manager, Supervisor, workers)
6. Pre-emptive priority (order of priority Manager, Supervisor, workers)</td><td>10</td></tr>
</table>

3	Consider a case study of your own and apply SJF and priority scheduling and calculate the average TAT, FT and WT. (Apply your own knowledge on CPU scheduling). Burst time will be the last three digits of your registration number thereby increase by five for the four processes. Note: If your registration number is 2127240501001 then the burst time for Process 1 is 001 and thereby increase by 5 for the Process 2, Process 3 and Process 4.			**20**
			Total Marks	**40**

SET 3

<table>
<tr><th>Q. No</th><th>Question / Task / Activity</th><th>CO*</th><th>RBT Level</th><th>Marks</th></tr>
<tr><td>1</td><td>II) The table given below lists the processes, corresponding Burst time, Arrival time and Priority. Perform CPU Scheduling for various scheduling algorithms, draw the process Gantt chart and calculate the following for each category. (Note: all time units given in nano seconds)

a. Finishing time
b. Waiting time
c. Turnaround time
d. Average waiting time
e. Average Turnaround time.
Category of scheduling algorithms:
FCFS
g. Without Arrival time
h. With Arrival time
3. SJF
a. Without Arrival time
b. Preemptive (With Arrival time)
c. Non-Preemptive (With Arrival time)
PRIORITY
d. Without Arrival time
e. Preemptive (With Arrival time)
f. Non-Preemptive (With Arrival time)
ROUND ROBIN (Time Slice – 3ns)
<table>
<tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr>
<tr><td>P1</td><td>0</td><td>12</td><td>2</td></tr>
<tr><td>P2</td><td>1</td><td>5</td><td>6</td></tr>
<tr><td>P3</td><td>10</td><td>10</td><td>1</td></tr>
<tr><td>P4</td><td>5</td><td>18</td><td>4</td></tr>
<tr><td>P5</td><td>9</td><td>3</td><td>3</td></tr>
<tr><td>P6</td><td>3</td><td>1</td><td>5</td></tr>
</table></td><td>2</td><td>AN</td><td>10</td></tr>
<tr><td>2</td><td>A Network mail delivery System schedules and delivers the packets in the ready queue. Implement FCFS, SJF, SRTF, Priority (Pre-emptive and Non-Preemptive), Round Robin (No. of Packet=1) Scheduling. Which Scheduling is best to increase the throughput.
<table>
<tr><th>System Name</th><th>No. of Packets</th><th>Time of arrival</th><th>Priority</th></tr>
<tr><td>A</td><td>5</td><td>0</td><td>1</td></tr>
<tr><td>B</td><td>2</td><td>1</td><td>2</td></tr>
<tr><td>C</td><td>10</td><td>2</td><td>1</td></tr>
<tr><td>D</td><td>3</td><td>3</td><td>3</td></tr>
</table></td><td></td><td></td><td>10</td></tr>
<tr><td>3</td><td>Consider a case study of your own and apply SJF and priority scheduling and calculate the average TAT, FT and WT. (Apply your own knowledge on CPU scheduling). Burst time will be the last three</td><td></td><td></td><td>20</td></tr>
</table>

	digits of your registration number thereby increase by five for the four processes. Note: If your registration number is 2127240501001 then the burst time for Process 1 is 001 and thereby increase by 5 for the Process 2, Process 3 and Process 4.			
			Total Marks	**40**

SET 4

<table>
<tr><th>Q. No</th><th>Question / Task / Activity</th><th>CO*</th><th>RBT Level</th><th>Marks</th></tr>
<tr><td>1</td><td>The table given below lists the processes, corresponding Burst time, Arrival time and Priority. Perform CPU Scheduling for various scheduling algorithms, draw the process Gantt chart and calculate the following for each category. (Note: all time units given in nano seconds)

a. Finishing time
b. Waiting time
c. Turnaround time
d. Average waiting time
e. Average Turnaround time.
Category of scheduling algorithms:
FCFS
g. Without Arrival time
h. With Arrival time
SJF
i. Without Arrival time
j. Preemptive (With Arrival time)
k. Non-Preemptive (With Arrival time)
PRIORITY
l. Without Arrival time
m. Preemptive (With Arrival time)
n. Non-Preemptive (With Arrival time)
ROUND ROBIN (Time Slice – 3ns)
<table>
<tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr>
<tr><td>P1</td><td>0</td><td>4</td><td>4</td></tr>
<tr><td>P2</td><td>1</td><td>6</td><td>2</td></tr>
<tr><td>P3</td><td>3</td><td>3</td><td>5</td></tr>
<tr><td>P4</td><td>5</td><td>7</td><td>1</td></tr>
<tr><td>P5</td><td>7</td><td>2</td><td>3</td></tr>
<tr><td>P6</td><td>8</td><td>5</td><td>1</td></tr>
</table></td><td rowspan="2">2</td><td rowspan="2">AN</td><td>10</td></tr>
<tr><td>2</td><td>Photographer has to deliver the photos to the customers on or before the due period. Which method is suitable for him to satisfy the customer needs if he follows FCFS, SJF, Priority, Round Robin (time slice=10 photos). Priority is based on due date.
<table>
<tr><th>Name</th><th>No. of Photos</th><th>Order of Arrival</th></tr>
<tr><td>Alice</td><td>100</td><td>1</td></tr>
<tr><td>Bob</td><td>20</td><td>2</td></tr>
<tr><td>Cathy</td><td>10</td><td>3</td></tr>
<tr><td>Rodalph</td><td>50</td><td>4</td></tr>
</table></td><td>10</td></tr>
<tr><td>3</td><td>Consider a case study of your own and apply SJF and priority scheduling and calculate the average TAT, FT and WT. (Apply your own knowledge on CPU scheduling). Burst time will be the last three</td><td></td><td></td><td>20</td></tr>
</table>

	digits of your registration number thereby increase by five for the four processes. Note: If your registration number is 2127240501001 then the burst time for Process 1 is 001 and thereby increase by 5 for the Process 2, Process 3 and Process 4.			
			Total Marks	**40**

SET 5

<table>
<tr><th>Q. No</th><th>Question / Task / Activity</th><th>CO*</th><th>RBT Level</th><th>Marks</th></tr>
<tr><td>1</td><td>The table given below lists the processes, corresponding Burst time, Arrival time and Priority. Perform CPU Scheduling for various scheduling algorithms, draw the process Gantt chart and calculate the following for each category. (Note : all time units given in nano seconds)

a. Finishing time
b. Waiting time
c. Turnaround time
d. Average waiting time
e. Average Turnaround time.
Category of scheduling algorithms:
FCFS
o. Without Arrival time
p. With Arrival time
SJF
q. Without Arrival time
r. Preemptive (With Arrival time)
s. Non-Preemptive (With Arrival time)
PRIORITY
t. Without Arrival time
u. Preemptive (With Arrival time)
v. Non-Preemptive (With Arrival time)
ROUND ROBIN (Time Slice – 3ns)
<table>
<tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr>
<tr><td>P1</td><td>0</td><td>7</td><td>4</td></tr>
<tr><td>P2</td><td>1</td><td>4</td><td>2</td></tr>
<tr><td>P3</td><td>2</td><td>6</td><td>5</td></tr>
<tr><td>P4</td><td>3</td><td>3</td><td>1</td></tr>
<tr><td>P5</td><td>6</td><td>5</td><td>3</td></tr>
<tr><td>P6</td><td>8</td><td>2</td><td>6</td></tr>
</table></td><td rowspan="2">2</td><td rowspan="2">AN</td><td>10</td></tr>
<tr><td>2</td><td>A machine center in a job shop for a local fabrication company has five unprocessed jobs remaining at a particular point in time. The jobs are labeled 1, 2, 3, 4, and 5 in the order that they entered the shop and their processing times are 11, 29,31, 1, 2. The respective processing times and due dates are given in the table below. Implement FCFS, SJF, Priority (Non-Pre-emptive), Round Robin (No. of processing days=5) Scheduling. Priority is based on increasing order of processing time.</td><td>10</td></tr>
<tr><td>3</td><td>Consider a case study of your own and apply SJF and priority scheduling and calculate the average TAT, FT and WT. (Apply your own knowledge on CPU scheduling). Burst time will be the last three digits of your registration number thereby increase by five for the four processes.</td><td></td><td></td><td>20</td></tr>
</table>

	Note: If your registration number is 2127240501001 then the burst time for Process 1 is 001 and thereby increase by 5 for the Process 2, Process 3 and Process 4.			
			Total Marks	**40**